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PROJECT DOCUMENT

Republic of North Macedonia

РЕПУБЛИКА СЕВЕРНА МАКЕДОНИЈА - REPUBLIKA E MAQEDONISË SË VERIUT
МИНИСТЕРСТВО ЗА ЖИВОТНА СРЕДИНА И ПРОСТОРНО ПЛАНИРАЊЕ - Скопје
MINISTRIA E MJEDISIT JETËSOR DHE PLANIFIKIMIT HAPËSINOR - Shkup

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Project Title: Tackling Air Pollution in the City of Skopje

Project Number: 00109164

Implementing Partner: United Nations Development Programme

Start Date: 12 December 2019 End Date: 31 December 2021 PAC Meeting date: 11.12.2109

Brief Description
Air pollution is a major cause of premature death and disease and is the single largest environmental health risk in Europe, causing around 400,000 premature deaths per year. The report released in June 2019 by the U.N. Environment Program and prepared in cooperation with the World Health Organization confirms that the population in the western Balkans is exposed to some of the highest concentrations of air pollution in Europe, up to five times higher than the national and the EU guideline levels. Moreover, the three cities – Skopje, Bitola, and Tetovo – were ranked in 2017 as among the top ten most-polluted in Europe with conditions worsening tangibly and measurably in over the past five years. The project aims to demonstrate a multi-pronged intervention to tackle air pollution in the City of Skopje linked to the residential sector. Its objective is to establish a fully functional platform that brings all traditional and non-traditional partners to work together to address the issue. The project has four main components: Component 1: Develop a comprehensive monitoring system for the pilot area, and a coordination platform to tackle air pollution; Component 2: Implement regulatory changes necessary to transitions towards a lower emission household energy system; Component 3: Demonstration of measures that address the causes of pollution for household heating, and Component 4: Build public awareness. The project shall enhance the coordination among all the responsible actors to ensure a "whole of government" and "whole of society" response to the air pollution challenge. The Ministry of Environment and Physical Planning and the City of Skopje will be the main project partners and beneficiaries, but the project shall peruse the involvement of all relevant stakeholders who can contribute to mitigate the problem. The mobilizing capacity of academia and civil society organizations shall also be tapped to devise solutions and help change public behaviours. The project shall support the advancement of the country's EU accession agenda and is aligned to and will use where possible locally owned systems to develop sustainable outcomes with a strong local ownership. At the same time, the project will facilitate the access to knowledge and experience from EU member states and advanced economies worldwide.

Contributing Outcome (UNDAF/CPD:

Outcome 4: "By 2020, individuals, the private sector and state institutions base their actions on the principles of sustainable development, and communities are more resilient to disasters and environmental risks"

Indicative Output(s) with gender marker: 2

Total resources required:	**2,056,766.76 $USD 20,000,000 SEK (UN Exchange Rate for November 2019 1$US = 9.724 SEK)**	
	UNDP:	**65,000 $USD (SEK 632,060.00) Parallel funding)**
	Donor (Swedish Government):	**20,000,000.00 SEK**
	Government:	
	In-Kind:	
Unfunded:		-

Agreed by (signatures):

Government	Government	Implementing Partner - UNDP
Naser Nuredini Minister of Environment and Physical Planning	Petre Shilegov Mayor	Narine Sahakyan Resident Representative
Date:	Date:	Date: 12. 12. 2019

I. DEVELOPMENT CHALLENGE

Air pollution is a major cause of premature death and disease and is the single largest environmental health risk in Europe (Lim et al., 2012; WHO, 2014, 2016a; GBD 2016 Risk Factors Collaborators, 2017; HEI, 2018), causing around 400 000 premature deaths per year. Heart disease and stroke are the most common reasons for premature death attributable to air pollution, followed by lung diseases and lung cancer (WHO, 2014). The International Agency for Research on Cancer has classified air pollution in general, as well as PM as a separate component of air pollution mixtures, as carcinogenic (IARC, 2013).

Both short- and long-term exposure of children and adults to air pollution can lead to reduced lung function, respiratory infections and aggravated asthma. Maternal exposure to ambient air pollution is associated with adverse impacts on fertility, pregnancy, new-borns and children (WHO, 2005, 2013b). There is also emerging evidence that exposure to air pollution is associated with new-onset type 2 diabetes in adults, and it may be linked to obesity, systemic inflammation, ageing, Alzheimer's disease and dementia (RCP, 2016, and references therein; WHO, 2016b).

Air pollution is not the largest environmental health risk only in Europe but it has become the single largest environmental health risk worldwide, with the WHO blaming harmful particulate matter (PM) for as many as 8.2 million deaths per year globally.[1] A recent study[2] has found that even "safe" levels of air pollution are linked to heart abnormalities similar to those seen during the early stages of heart failure. This global challenge is reflected locally in Macedonia, where three cities – Skopje, Bitola, and Tetovo – were ranked in 2017 as among the top ten most-polluted in Europe[3] with conditions worsening tangibly and measurably in over the past five years.

The most recent report of the European Environmental Agency "Air quality in Europe – 2018 report" that presents an updated overview and analysis of air quality in Europe from 2000 to 2016 reveals that the country has the highest PM10 concentrations in relation to the daily limit value in 2016, and it is second worst in Europe for PM2.5 concentrations in relation to the annual limit value in 2016.

Preliminary findings from the 'Air Pollution and Human Health: The Case of the Western Balkans report (UN Environment, June 2019) shows that the sum total number of premature deaths directly attributable to air pollution in the cities is nearly 5,000 a year. i.e. air pollution is directly responsible for up to one in five premature deaths in 19 Western Balkan cities. On average, people living in the Western Balkan cities studied lose up to 1.3 years of life to air pollution. The main sources of particulate matter emissions are thermal power plants that use lignite coal and household heating. Average concentrations of PM2.5 particulate matter in all but one of the 19 cities studied exceeded the World Health Organisation guideline level of 10 µg/m3. A daily PM10 limit of 40µg/m3 set out under national legislation was found to be exceeded between 120 and 180 days a year - especially during winter. In comparison, European Union member states are not permitted to breach this level for more than 35 days a year[4].

[1] http://www.who.int/mediacentre/news/releases/2016/deaths-attributable-to-unhealthy-environments/en/

[2] https://www.qmul.ac.uk/media/news/2018/smd/new-research-links-low-levels-of-air-pollution-with-serious-changes-in-the-heart-.html

[3] https://www.theguardian.com/cities/datablog/2017/feb/13/most-polluted-cities-world-listed-region

[4] https://www.unenvironment.org/news-and-stories/press-release/air-pollution-responsible-one-five-premature-deaths-19-western

A series of studies conducted under the leadership of the Ministry of Environment and Physical Planning with different partners has helped to identify the main causes. An assessment[5] of air quality in the 2005-15 period was prepared with support from the Finnish Meteorological Institute as part of an EU twinning project, which in 2016 also produced a detailed *Air Quality Improvement Plan for Skopje Agglomeration.*[6] This plan was adopted by an inter-sectoral group led by the Deputy Minister of Environment and Physical Planning.

According to this study, the most critical pollutant in Skopje is the suspended particle matter, of which PM_{10} concentrations exceed both the daily and annual limit values. On certain days, the hourly concentrations of PM_{10} particles[7] in Skopje were reaching 20 times more than maximum allowable levels. While particular days had high measurements of particulate matter, the averages have also been high:

- The average annual concentration of $PM_{2.5}$ in 2015 for Centar Municipality was 40.14µg/m3, and for Karposh Municipality it was 50.51µg/m^3 – versus an average annual maximum allowable concentration of $PM_{2.5}$ up to 25 µg/m^3 under the national law on air quality.
- The average concentration of PM_{10} in these two municipalities were 72.82µg/m^3 (Centar) and 57.06µg/m^3 (Karposh) - versus the average annual maximum allowable concentration of PM_{10} up to 40 µg/m^3.

The Finnish Meteorological Institute's study also found that air quality in Skopje has a seasonal trend, with PM_{10} and $PM_{2.5}$ concentrations higher during the heating season from October to April. The sector most responsible for emissions of PM is household heating. In particular, household (residential) wood burning is responsible for almost all (90%) of the primary PM emissions in Skopje. However, the emission amounts do not directly translate to an equal amount of pollution in the air, as the formation of pollutant concentrations is influenced by the height and location of the emission release. According to the study, heating through wood burning contributes to approximately 30% of the particulate concentrations. Furthermore, air pollution within households from inefficient stoves can be expected to have a more significant impact on women and children, depending on hours spent indoors.

To better understand this issue, UNDP conducted a study of household heating practices in Skopje in 2017.[8] This study surveyed 5,044 households in the 17 municipalities in and around Skopje and showed that only 21% of residents were connected to the district heating (DH) system, while 45% used woodstoves to heat their houses. Inefficient wood-burning stoves are one of the most polluting forms of heating. This pattern applies to urban as well as rural areas. Each year, Skopje residents burn 317,000 tonnes of wood (or an amount equal to all the trees on Vodno Mountain). On average these heating systems operate 13.5 hours a day on working days during the heating season.

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[5] http://air.moepp.gov.mk/wp-content/uploads/2017/07/AirQualityReport_EN.pdf

[6] https://eeas.europa.eu/sites/eeas/files/activity_3.2_benchmark_3.2.1_skopje_air_quality_improvement_plan.pdf

[7] PM_{10} is short for particulate matter which is 10 micrometers or less in diameter, $PM_{2.5}$ is particulate matter which is 2.5 micrometers or less in diameter.

[8] The results are available at http://skopjesezagreva.mk

Most of those using firewood for heating use inefficient and old furnaces. Another particularly worrying finding is that 40% of households admit to burning plastic, garbage and other toxic materials for heating. This practice is particularly pronounced in low-income and Roma neighbourhoods (98% of Roma heat with woodstoves, and many of them burn toxic materials).

In rural areas, mostly firewood is used for heating and more than 85% of households use it. The remaining households mainly use electricity for storage heaters and electric boilers. A smaller percentage of households use heat pumps.

In urban areas:

- 38.7% of households use electricity (more or less efficient, depending on the system);
- 29.8% are connected to the DH system;
- 28.2% use firewood; and
- A small percentage use other fuels (not specified in the study).

In addition to inefficient heating sources, the residential building stock is also very inefficient. In Skopje, 51% of households reported having no insulation, while another 42% have only partial insulation.

This study also helped in understanding the correlations between the households' heating choices and their financial status. The main criterion in the selection of a heating system among citizens is financial (investment and monthly operating cost) – described as important for 74% of respondents. Only 1.3% said that when choosing a heating system, they were guided by how much such the system pollutes the air, suggesting that there is a very low public awareness or concern about the dangers of air pollution and how household heating systems impact air quality. The problem with heating choices is not solely related to household income, where 30% of the households in rural areas would like to use firewood for heating' regardless of the heating cost. The study suggests that the use of firewood does not depend on the household's financial situation, but on household members' education level.

These findings resonate with another recent report about the biomass use for heating in Macedonia (EBRD-WB, 2017), which highlights that biomass meets 39% of current heat demand, with most of it being inefficient (74% of biomass is used in traditional, inefficient stoves, and not properly dried)[9]. It is the end user's heating behaviour (fuel selection, stove operation and maintenance), which is most decisive for achieving high efficiency and low emissions.

The issue with air quality, in addition to being covered by several national and international media sources, has been also raised by civil society organisations, with concerns for lower income households. According to the report by UNDP, the high unemployment rate and low living standards contribute to inappropriate heating choices in households. People are forced to buy cheap fuel for heating their homes, greatly affecting pollution levels.

It should also be noted that the widespread use of inefficient wood heating furnaces results indirectly in pressure on forest resources and increased costs to households for energy with less levels of comfort. Therefore, the situation with residential heating indirectly leads to the existing problem of widespread illegal logging, including in national parks and other protected areas.

[9] International Bank for Reconstruction and Development / The World Bank (2017). Biomass-Based Heating in the Western Balkans - A Roadmap to Sustainable Development.

Meanwhile, more environmentally effective sources of heating are available including most notably:

- District Heating based on cogeneration from a thermal (natural gas-fired) power plant – meaning that effective emissions (GHG and otherwise) are zero for the use of waste heat.
- More efficient wood heating units which can heat more area of a dwelling utilizing less wood and resulting in significant reductions of particulate matter and other pollutants (e.g. carbon monoxide).
- Thermal air-sourced and ground-sourced heat pumps which – while expensive – can be popular and cost effective.

A number of underlying factors have led to this current situation which need to be addressed in order to tackle air pollution from domestic heating sources. These include the following:

- **Inadequate monitoring of air quality and potential sources of pollution.** Though initial studies have been carried out and the general sources of pollution have been identified, Skopje lacks a comprehensive monitoring system that informs policy makers, private sectors, investors and the public to understand the 'real costs' of pollution. As mentioned in previous studies[10], reliable air quality monitoring data was not available[11] to exhaustively complete the source appointment studies, and measurements did not fulfil the minimum data coverage requested by legislation, resulting in the unavailability of reliable annual average concentrations.
- **Insufficient coordination amongst stakeholders.** A number of stakeholders are involved or could be involved in addressing this issue, including the local government, national government, various donor organisations, and community groups. At the moment there are not effective platforms for coordinating action.
- **Inadequate regulatory framework.** Notably this is related to standards for heating units to ensure they are less polluting or more efficient – which would comply with the EU directive on ecodesign[12]. It also relates to the regulatory framework governing the district heating use and distribution, which is insufficiently developed to encourage the use of the district heating grid.
- **Lack of successful examples that demonstrate results.** This relates to a lack of neighbourhood - or building-level examples of switching to district heating systems or improving end-use efficiency of buildings.
- **Large number of inefficient buildings with no or limited insulation, and few workable schemes available to owners to finance or implement retrofitting.** This results in more fuel (wood, plastics) being burned to heat space to limited comfort.
- **Limited availability of efficient heating technologies.** While the district heating system is mostly available in Skopje, in some parts of the City it is too expensive to connect to. In these cases, it is necessary to have efficient wood furnace technology available at a reasonable cost, and to work with district heating providers to improve affordability and residential uptake.
- Related to this, as described above, **inefficient wood-burning stoves are used for heating** predominantly. These should be phased out of the market entirely.
- And similarly, there is **low awareness among the retailers of the key role that they can play in the low-carbon economy chain** and the contribution to the behavioural change by

[10] http://air.moepp.gov.mk/wp-content/uploads/2017/07/AirQualityReport_EN.pdf

[11] For $PM_{2.5}$

[12] https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=celex%3A32009L0125

offering smart and affordable packages to the households (focusing on low-income ones) to speed up the process of replacement of old inefficient stoves.

- **Low awareness levels about the sources and effects of air pollution among certain categories of citizens**, as relates to the impact of inefficient burning on health and the environment, and the possibilities for more cost-effective heating sources / stoves.
- **Widespread polluting behaviours** such as open burning of wood and plastics, as well as illegal logging.

Relevant European Union Legislation

The EU has been working for decades to improve air quality by controlling emissions of harmful substances into the atmosphere, improving fuel quality, and integrating environmental protection requirements into the transport, industrial and energy sectors.

Ambient air quality standards are set out in the *Ambient Air Quality Directives* (EU, 2004, 2008), and they require the Member States to adopt and implement air quality plans and meet standards in order to protect human health and the environment.

National emission reduction targets established in the National Emission Ceilings (NEC) Directive (EU, 2016), requiring Member States to develop National Air Pollution Control Programmes by 2019 in order to comply with their emission reduction commitments.

Emission and energy efficiency standards for key sources of air pollution from vehicle emissions to products and industry. These standards are set out in EU legislation targeting industrial emissions, emissions from power plants, vehicles and transport fuels, as well as the energy performance of products and non-road mobile machinery.

The Seventh Environment Action Programme, 'Living well, within the limits of our planet' (EU, 2013) recognizes the long-term goal within the EU to achieve 'levels of air quality that do not give rise to significant negative impacts on, and risks to, human health and the environment'. To meet this goal, effective air quality policies require cooperation and action at global, European, national and local levels. In line with the principle of subsidiarity, policies must be developed at national, regional and local levels, implementing measures tailored to specific needs and circumstances.

The Clean Air Programme for Europe (CAPE), published by the European Commission in late 2013 (European Commission, 2013), aims to ensure full compliance with existing legislation by 2020 at the latest, and to further improve Europe's air quality, so that by 2030 the number of premature deaths is reduced by half when compared with 2005. Following the proposals made in this context, the NEC Directive was revised and approved in 2016 and a new directive (EU, 2015) on the limitation of emissions of certain pollutants from medium combustion plants was approved in 2015.

The Ecodesign Directive (2009/125/EC) and associated regulations adopted by the EU set minimum performance standards for energy consumption and associated emissions from various household energy consuming products – notably solid-fuel burning stoves and boilers. Macedonia has adopted some of the relevant regulations but not those related to solid-fuel burning heaters.

The Energy Efficiency Directive (2012/27/EU) and Energy Performance in Buildings Directive (2010/31/EU) which outline requirements for developing targets for energy savings, evaluating the energy consumption of buildings, and meeting minimum energy performance standards in newly constructed or renovated buildings.

In addition, the EU is supporting and facilitating the Member States to take the measures necessary to meet their targets, and the enforcement action to help ensure that the common objective of clean air for all Europeans is achieved across the EU. This includes the EU Urban Agenda and Urban Innovative Actions, which facilitate cooperation with and among city actors to address air pollution in urban areas across the EU. The CAPE also envisages a regular update of the impact assessment analysis, to track progress towards the objectives of the Ambient Air Quality Directives.

European Union Accession

The Programme of the Government 2017 - 2020 declares its unbinding political will to establish a reform-oriented government focused on country's common strategic priorities, accession to NATO and the European Union. In its Programme, the Government commits to provide better quality of environmental media and protection of natural resources through non-corrupt effective institutions; strengthened, equipped and professional inspection services; private sector support, so it can meet the environmental protection standards. In this context, they promise to provide conditions for permanent air quality monitoring and monitoring of air, water and soil emissions, and to make these data available to the public, science community and civil society. The Government also obliges to establish dedicated environmental health indicators for assessing risks to children's health from air pollution.

However, the success in this regard can only be attained by a visible reform agenda. The latest EU Progress Report 2019 issued on 29 May 2019, identifies number of deficiencies at the level of preparation in the areas of Environment and Climate Change (Chapter 27). The report recognizes some progress has been achieved in further aligning policies and legislations with the acquis, particularly in water, nature protection and waste sectors. However, it also highlights that significant efforts are needed about implementation and enforcement. Administrative capacity at central and local level remain weak and insufficient. Cooperation with civil society has improved, but further efforts are needed for effective public participation and consultation in decision-making processes. The implementation of Environmental Impact Assessment and public consultations need to improve, especially at the local level. No progress has been made in adopting the Law on Environmental Inspection and the Environmental Liability, INSPIRE and Environmental Crime Directives are still not fully aligned and implemented. In the coming year the country should in particular: implement air quality improvement measures by ensuring efficient coordination between central and local authorities, and the allocation of sufficient financial resources; implement the Paris Agreement by developing a comprehensive strategy on climate-related action, consistent with the EU 2030 framework and start the process of developing a National Energy and Climate Plan, in line with Energy Community obligations.

On ***air quality***, the Progress Reports determines that although the legislative alignment is almost complete, its implementation remains weak. The lack of sufficient administrative and financial resources is still the main obstacle to putting in place air quality improvement measures. Air pollution levels are high in major cities. Insufficient human, technical and financial support for the national air quality monitoring network resulted in several interruptions. A special allocation to fund air quality improvement measures has been introduced in the 2019 state budget for the first time.

II. STRATEGY

Effective action to reduce air pollution and its impacts requires a good understanding of its causes, how pollutants are transported and transformed in the atmosphere, how the chemical composition of the atmosphere changes over time, and how pollutants impact humans, ecosystems, the climate and subsequently society and the economy. To tackle air pollution, collaboration and coordinated action at national and local levels must be sustained, and it must be backed up with adequate environmental, climate and sectoral policies. Holistic solutions involving technological development, structural changes and behavioural changes are also required, together with an integrated multidisciplinary approach.

Taking into consideration that the heating practices in the residential sector are the main contributor to the air pollution of the City of Skopje, as well as the availability of the current funding, the approach for this project is to carry out a multi-pronged intervention to tackle air pollution linked to the residential sector. It is aimed to establish a fully functional platform that brings all traditional and non-traditional partners to work together to address the issue.

In the face of the multiple and intersecting causes of that contribute to Skopje's air pollution problem, any program aimed at providing answers must resist the temptation to satisfy public pressure for action by pursuing high-profile but partial or short-term solutions. This is a true dilemma given the seriousness of the health threat facing Skopje residents and their understandable impatience for decisive action. Fortunately, there are some obvious "no regrets" measures that can be undertaken or at least tested in pilots.

Given the complexity of the causes, there is also an acute need for coordination among all the responsible actors to ensure a "whole of government" and "whole of society" response to the air pollution challenge. The City of Skopje and other municipalities have so far taken the lead, but full engagement by central-level authorities is also essential given the disparate sources of pollution and dispersed regulatory responsibilities. Ministerial engagement needs to extend well beyond the Ministry of Environment and Physical Planning to other line ministries, and the project shall peruse the involvement of all relevant stakeholders who can contribute to mitigate the problem. The mobilizing capacity of civil society organizations shall also be tapped to devise solutions and help change public behaviours.

In designing the interventions, system thinking method, with the citizens in the centre of the solution (human-centred design) will be used. The system-solving approach will aim to explore a roadmap, that:

- Have clearly identified specific target groups
- Is focused at local level - designing and testing local actions specific for municipality or neighbourhood in Skopje Valley
- Relies on non-traditional forms of partnerships with private sector and academia
- Involves every single institution that have a roles and responsibilities in addressing the system
- Is focused on changing citizens' behaviours (incentives vs. normative)

The project team will utilize the capacity and the network of Skopje Lab – the City of Skopje Innovation Lab, established in 2017 by the City of Skopje and UNDP as an experimental space for co-designing and testing new approaches for solving Skopje's major urban challenges.

It will bring about change in the behaviour of people, and in the use of energy in Skopje through a comprehensive approach involving:

- Effective and inclusive monitoring / understanding of the sources of pollution

- Changes in the regulatory framework to bring about changes in behaviour amongst market operators and consumers
- Investments in interventions that will reduce energy consumption and pollution – including particulate matter, CO, and GHGs
- Communications / awareness raising activities to ensure public buy-in and better understand how to adjust programmes to achieve lasting impacts.

The project also plans to engage NGOs through competitive open call for grants or services:

- NGO that would work with the potential beneficiary households in Lisice;
- NGO for awareness raising/advocacy activities;
- NGO for events organization (hackathon and data visualization).

Additionally, for the energy audits, efforts will be made to closely collaborate with the Association of Energy Auditors as a specialised association of citizens.

This approach will integrate gender considerations into each relevant activity – including ensuring the disaggregation of data gathered on impacts, encouraging gender-balanced participation in project activities, and specifically researching any gender differences as relates to air pollution impacts.

The intended project outputs and results will contribute to achieving the national priorities but also the outcomes set in the UNDP Country Project Document 2016-2020 signed by the Macedonian Government. It will also contribute to the advancement of the following SDGs targets 3.9, 7.1, 9.4, 11.6, and 13.2.

The national objective to become an EU member is the main driving force for the project and an important factor for the sustainability of project results.The relevance of the project stems from the national objective to apply EU regulations in the quest to become an EU member. The quotations above from the most recent EU report on accession indicate that there are bottlenecks in this process, and the project shall directly contirbute to overcoming the identified deficiencies related to the air pollution.

The project is aligned to and will use where possible locally owned systems to develop sustainable outcomes with a strong local ownership. At the same time, the project will facilitate the access to knowledge and experience from EU member states and advanced economies worldwide. The examples and experience form Sweeden will be of particularly interest (eg. https://smartcitysweden.com).

The project funding would help to lay sturdy foundations for the effective planning and mobilization of other donor funding and government cost-sharing and, to the extent possible, other forms of development finance from public and private sources, including innovative instruments. The project would serve as a point of orientation for other donors seeking to support a comprehensive, credible plan of systemic solutions to air pollution.

UNDP would structure the project to efficiently pool funding from multiple sources while providing the donors who need it with high visibility and ensure that scarce resources are not wasted on partial or temporary solutions. It would also provide leverage for future larger-scale grant and loan packages available from the Global Environmental Facility (GEF), EBRD, the European Investment Bank and the Council of Europe Development Bank.

Through parallel funding UNDP shall implement a complementary project called "If we were counting" that is part of the innovative initiatives funded by the City Experiment Fund (UNDP managed fund financed by the Slovak Government). The project is a real-time governance test

case that explores what could happen if we used algorithms to automatically regulate and fine heavy polluters and give that money back to residents to decide how to use it toward improving livelihoods in Skopje. It aims to incentivize government and industry to work on new solutions to decrease air pollution, bring transparency to corrupt processes, decrease apathy, and increase trust in government. By using geospatial data and sensors, algorithmic regulation and participatory budgeting, the project aims to design an integrated test platform that could hold government and industry accountable for real-time regulation of GHG emissions and incentivize residents' participation in reducing air pollution.

III. RESULTS AND PARTNERSHIPS

Expected Results

The project is structured around four main components with sub-components and activities described below.

Component 1: Develop a comprehensive monitoring system for the pilot area and a coordination platform to tackle air pollution

Building on the research, analysis and planning already conducted by the Ministry of Environment and Physical Planning and the City of Skopje, UNDP will work with local stakeholders to create a framework for the long-term transformation of both key economic sectors and human behaviours. Such a comprehensive and robust planning effort will be required to shift the responses from ad-hoc interventions and quick fixes focused on symptoms, to a long-term transformative process that will enable society to make optimal choices among different investment and growth opportunities.

This component will involve the following sub-components and activities.

Sub-component 1.1: Development of a comprehensive monitoring system for the pilot area

Activity 1.1.1 Improvements to the existing monitoring system to measure air pollution by extending the network of monitoring stations and setting up an IoT and machine learning-based IT platform

Under this activity, a comprehensive monitoring system for indoor and outdoor air pollution will be implemented through the improvement and upgrade of existing monitoring stations in in the selected strategic pilot area of the Skopje agglomeration. The automated stations will determine SO2, CO, NO and ozone levels. Flowing aerodynamic diameter particles of less than 10, 2.5 and 1.0 microns (PM10, PM2.5, PM1.0) are determined gravimetrically. Particles of PM10 are defined as Pb, Cd, Mn, As, Ni, Zn, Cu and Fe, polycyclic aromatic hydrocarbons and anions of sulphates, nitrates and chlorides. The monitoring stations will be provided with an "Internet of Things" (IoT)/ machine learning-based IT platform.

Activity 1.1.2 Develop a monitoring system and an online platform for collecting and visualising data

In addition to fixed monitoring stations, a citizen-led monitoring system and an online platform for collecting and visualising data will be developed following successful examples of participatory mapping. Activities related to collecting and communicating citizens' data about heating practices will be used and expanded upon to capture on-going monitoring activities, providing a crowd-sourced pool of information. This platform should support other awareness and communication activities, ultimately helping society to understand how their behaviours affect air quality.

Activity 1.1.3 Continued monitoring of the contributions to the air pollution by repeating heating study at the end of the project to include both small companies and households

In the last phase of the project, continued monitoring of the contributions to air pollution will be implemented by repeating the study about heating practices, expanded by inclusion of small businesses in the research. The results from this activity will support decision makers and citizens to better understand how changes in air pollution levels are linked to their heating choices.

This sub-component shall be closely coordinated with the cooperation project between the Swedish Environmental Protection Agency (Swedish EPA) and the Ministry of Environment and Physical Planning (MoEPP) on Air Pollution, which overall objective is to support the preparation for EU accession in the environmental sector.

Sub-component 1.2 Modelling of air pollution sources

Activity 1.2.1 Source apportionment modelling - Identify specific geographical areas in which investing in mitigation measures will have the greatest impact

Information on pollution sources is essential to the design of air quality policies and source apportionment is required for the implementation of the EU Air Quality Directives. Under this component, modelling activities will identify specific geographic areas and economic sectors in which future legislation and investments should concentrate in order to achieve the greatest impact.

This sub-component will be carried out with the support from local educational and scientific institutions i.e. the University "Goce Delcev", Stip

Sub-component 1.3 Implementation of a coordination platform

Activity 1.3.1 Learning exchange and linkages with other initiatives to inform a long-term plan of action - Convening stakeholders, donors and relevant actors to implement necessary changes identified

The monitoring and modelling activities will provide the basis for an informed plan of action, coordinated by UNDP, who will act as enabler of a coordination platform that links together multiple stakeholders. This coordination platform will have the purpose of convening donors, government actors, CSOs and private sector stakeholders to implement necessary changes identified through the source apportionment studies, other relevant studies/plans and community-based activities. The online platform will be a space of interaction involving the different stakeholders who share a common interest and commitment to tackle the air pollution and interact to improve common understanding, build confidence, define roles, and agree on joint action. It can facilitate dialogue between the Government, civil society as a representative of citizens, private sector in identifying the bottlenecks and participating in the seeking of consensual solutions that will tackle the problems on medium and long term. In addition, a long-term action plan will be also informed by learning exchange activities with other initiatives currently being implemented in Europe. This will create linkages with stakeholders and communities tackling similar problems in other countries. Experiences with low-emission districts in Sweden and with citizen-led monitoring systems could be regarded as examples.

During the inception phase of the project the shape of the platform shall be agreed. In this context, several key issues will be taken into consideration: *sustainability of the platform* (how to create a

dynamic within the platform which may mobilize funding in order to set up joint actions aligned with the common vision), *ensuring cohesion* among stakeholders that will be part of the platform (particularly group facilitation and management given that different types of entities might join the platform; managing possible power imbalances among the different entities in the platform); *inclusion of stakeholders* (the platform must include all stakeholders and ensure that no one who is interested to join is excluded; also it has to attract the private sector and interested donors who are willing to support the efforts to address the air pollution on a long run).

In summary, the results achieved under Component 1 will include:

- The development of a comprehensive monitoring system for the pilot area achieved by upgrading the current network, by developing an online platform to collect and visualise data, and by repeating the heating study;
- The modelling of air pollution sources through a source apportionment study;
- The implementation of a coordination platform which will creates synergies and linkages with other initiatives to inform a long-term plan of action.

Component 2: Implement regulatory changes necessary to transitions towards a lower emission household energy system

This component will support the MOEPP to implement legislation changes that are necessary to transition towards a lower emission household energy system. These will include necessary requirements in building regulations, particularly Eco design Directive.

Activity 2.1.1 Assistance to support legislation changes - ecodesign adoption and implementation to eliminate poorly performing wood stoves / solid fuel heaters

The Ecodesign Directive establishes a framework to set mandatory requirements for energy-using and energy-related products, forcing manufacturers to reduce the energy consumption and other negative environmental impacts of products at the design stage. In the case of solid fuel heaters, the ecodesign regulations also set minimum performance standards (maximum pollution values) for various types of pollution (particulate matter, CO, etc.) This activity will support the Government in transposing the relevant pieces of secondary legislation for solid fuel heaters. The introduction of these changes in the regulatory framework will lead to the reduction of air pollution by forcing market operators to only place on the market new efficient stoves – resulting in the phase out of old, dirty, inefficient ones.

This activity will be closely coordinated with other ongoing projects, particularly within IPA, that are supporting the EU approximation process.

Activity 2.1.2 Assistance to support legislation changes - waste management (promote recycling of plastic)

This activity will include working closely with the local and national government in developing appropriate regulations to tackle waste management problems and to encourage recycling to the extent that it is not covered by other complementary projects on waste management implemented by the Ministry of Environment and Physical.

Activity 2.1.3. Regulatory impact assessment - cost-benefit, social, environmental, and gender impact for regulatory changes

A regulatory impact assessment to establish impacts and role players for these policies, together with training / awareness raising activities on the impact of these changes for relevant government officials will be carried out under this component.

Component 3: Demonstration of measures that address the causes of pollution for household heating to show proof of concept

This component includes a series of high-impact area-based measures that address some of the previously-identified priority interventions and demonstrate a *low-emission district concept* with an aim to contribute to medium-term changes in the way the air pollution is tackled in the country.

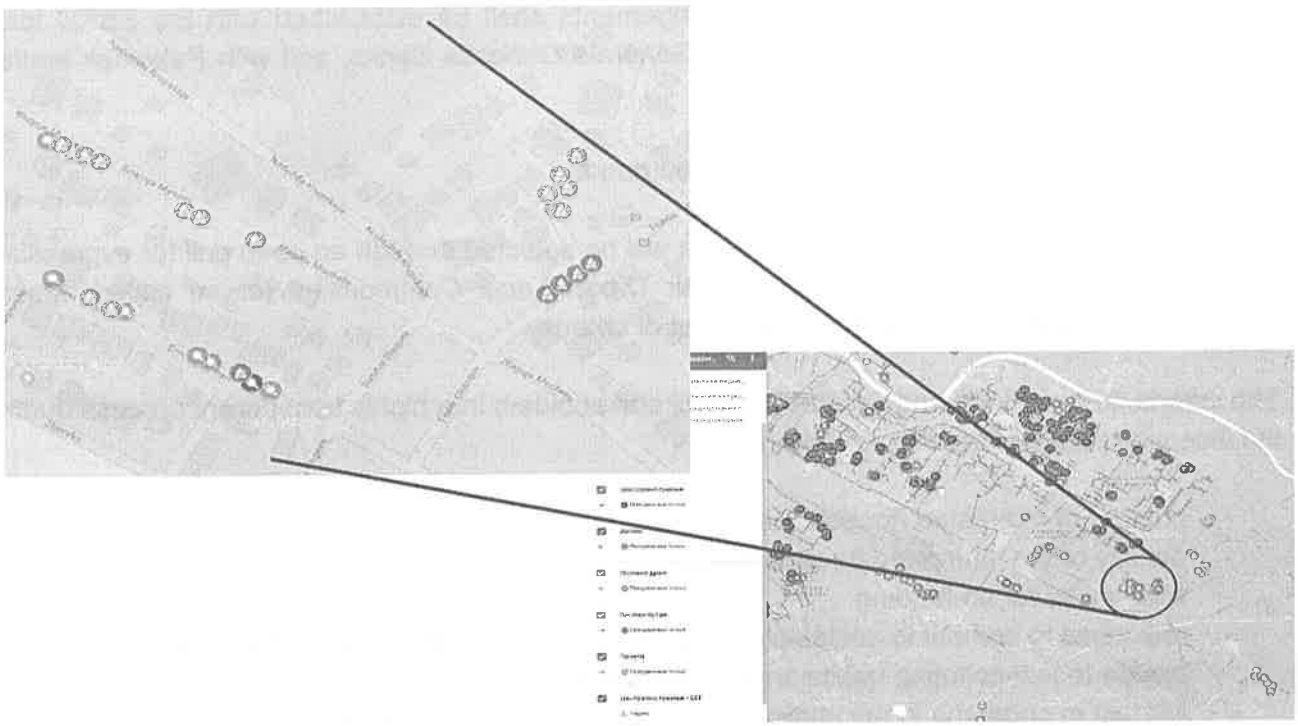
This component will be delivered through a subsidy programme for low-emission energy efficient districts

With the goal of reducing emission through energy-efficiency improvements in residential buildings, households in the targeted area will be supported in implementing thermal rehabilitation, efficient heating and sustainable waste management solutions. The scope and the volume of the activities will depend on availability of funding and it can be extended) if new funding is mobilized in the course of the project implementation. The specific area of the city is proposed to be targeted to ensure a cohesive intervention rather than having buildings in various parts of the city undergoing interventions. Furthermore, the activities will focus on specific geographical area to encourage a "district-wide" concept of emission reduction activities and therefore (a) reduce the energy consumption and pollution in a noticeable geographic area and (b) increase the awareness raising impact of the interventions.

Based on the undertaken analysis and some field exploration as well as expressed interest of the government authorities at all levels (Ministry of Environment and Physical Planning, City of Skopje), the area of Novo Lisice (neighbourhood of individual residential buildings) is proposed as suitable location for this intervention.

For many years now, Novo Liscie has the worst air quality in the whole valley. According to the last available official data of the Ministry of Environment, in 2016, the average daily concentrations of PM10 have exceeded the limits for 144 days. The geographical location of Novo Lisice and its lowest altitude in the Skopje valley keep the polluted air longer. The winds are not favourable either.

The underlying causes are numerous, while the source of pollution is predominantly burning firewood. Over 30% of all households in Municipality of Aerodrom use firewood for burning, with almost all of them in individual residential buildings, and over 90% in the area of Lisice. The usage of firewood, but also all other highly polluting waste materials such are tyres, polished wooden furniture, plastics, used car oil etc. is the major polluter in this area. This does not only cause only very bad quality of ambient air pollution, but also more harmful indoor air pollution.



Picture 1: Geographical distribution of the household heating in Aerodrom and a neighbourhood in Lisice

The vulnerable groups are disproportionately affected by the polluted air. Children, elderly and non-working women are particularly affected due to longer hours exposure to the indoor air pollution. Over 41% of the households in Aerodrom that use firewood have children under 18.

The usage of firewood, as found in the household heating survey cited above, is not particularly corelated to the monthly income, but shows high correlation with the educational level of the respondent of the survey. This area is populated with residents with relatively low monthly income, and lower levels of education, predominantly with completed high school and even elementary school.

This area has no connection to any central heating system, nor gas supply. While the gas might be brought to the area, the secondary and tertiary gas network process might take 5 or more years, and the whole process will be costly and long, and residents not interested to connect to, without major subsidies from the state, which is another inefficient model.

In their strategic plan, the central heating provider, following a detailed cost-benefit analysis which proved as expensive and inefficient approach, decided not to extend the network in this area.

Based on this analysis, it is proposed to implement area-based low-emissions district approach based on efficient and renewable energy sources, highly energy efficient homes and sustainable waste management solutions.

The following activities will, therefore, be undertaken under this sub-component:

Activity 3.1.1. Establishment of partnerships with all relevant stakeholders from private and civic sectors and formalization of the partnership with the Municipality of Aerodrom.

Under this activity, partnerships will be established or formalized with the identified partners. The partnership with the Municipality of Aerodrom will be formalized through signing a Memorandum of

Understanding, and similar partnership arrangements shall be established with the EBRD loan disbursing banks – Sparkasse and Société Générale Ohridska Banks, and with Pakomak and/or waste recycling licenced companies.

Activity 3.1.2 Selection of intervention neighbourhood

The most appropriate neighbourhood in Lisice will be selected through an open call for expression of interest by the communities through their Urban Local Communities (or so called "Mesni Zaednici"). The site will be selected against set of criteria.

The criteria, to be refined and validated with all stakeholders in a highly transparent process during the inception phase of the project, will include:

- number of interested households
- density of the interested households
- willingness for co-funding
- readiness to commit to sustainable maintenance of the newly installed equipment and pledge to low-polluting habits and technologies.
- interest in engaging in the implementation of the measures (in the available timing)
- readiness to go through an educational curriculum on sustainable and low-carbon communities
- share of vulnerable population, children, elderly, non-working parents, people with disabilities
- income averages and share of families' beneficiaries of social assistance
- share of houses or parts of houses that are illegally built or have not completed the process of legalization yet.

Civil Society Organization will be selected at open call to engage with the communities and support them in the process of mobilizing, education and awareness raising, as well as in the application process.

Following the selection of the neighbourhood, the CSOs will continue working with all households to complete the documentation process and monitor the progress of design and implementation of the measures.

Activity 3.1.3. Energy audits and co-design of energy-efficient measures, less-polluting heating systems and yard/area improvements with selected stakeholders

Once the neighbourhood is selected and letters of intent to participate in the programme are signed with all households, team of certified energy auditors and other experts (to be determined based on the needs) will be selected to undertake an energy audit of each single house to assess the household needs and existing situation, i.e. the wall and roof insulation, heating systems and behavioural patterns, as well as waste storage and disposal and yard condition.

The team will then develop an individual proposal for each house of what is required to ensure the household decrease its emissions to the lowest possible levels. The new heating technologies will be selected based on the most economical and ecological criteria, and it can include solar/photovoltaic combined with other systems, pellet stoves or inverter Air Conditioners.

Audit methodology and tools to support the process will be developed by team of relevant experts and reviewed and validated with the households.

Once the individual model and financial estimates are completed, the audit teams will present the funding options to the households and jointly identify the best possible funding sources.

Activity 3.1.4 Financing of interventions: subsidy for thermal rehabilitation for individual houses combined with redesign / installation of new heating elements, rubbish removal and yards arrangements where needed

The measures will be implemented through a range of support mechanisms designed for different types of customers.

The funding model will be refined and validated in co-design workshop at the very beginning of the project. The basic funding framework will be consisted of:

- Subsidy
- Household co-funding

The subsidy will be covered by the project funding and/or by the Municipalities of Aerodrom and/or by City of Skopje. For example, all households that will need to replace the heating appliances with pellet stoves or air conditioners, shall apply for the subsidies of the City of Skopje.

The subsidies covered by the project funding will range from 30 to 70% of the total value, with 100% on exceptional basis for most socially-deprived families. The subsidy per household will not exceed $10,000. The amount of the subsidy will be determined based on previously developed methodology by international and local experts' team, validated with the households and transparently published.

Some of the criteria will include:

- Number of members of the household
- Average monthly income of all members
- Number of children and students and member with disability in the household
- Health conditions of the members and the length of exposure to the indoor pollution
- Their investment in the renovation of the house in the last 5 years
- Other criteria as suggested by the experts' team or the residents

The subsidies will not be disbursed to the households, but UNDP through public procurement process will select the best offerors for each of the required construction or retailers' services. This will ensure transparent and accountable process with best value for the money invested.

The owners' co-funding part will secure the remaining of the budget. This can be either owner's investment, EBRD loan (which with the grant component and low interest rates is affordable), or through tailored-made loan programmes with the commercial banks that will be negotiated with the banks.

In the implementation of this activity, UNDP will closely work with the team of the EBRD energy efficiency programme which is implemented in partnership with Ohridska Banka Société Générale and Sparkasse Bank. The EBRD through the two banks provides loans to households for energy efficiency improvements in their homes, so joint approach is agreed in the project area where UNDP will focus its interventions.

During the process of refurbishment and creation of the low-emission district, participating households will undergo a series of trainings and educational activities, through traditional and non-traditional mechanisms, described in the last component below.

Once completed, the beneficiaries of the programme will sign binding contracts for residential low-emissions contributions. The resident monitoring system that will be designed will include component for distributed monitoring responsibility among the residents who will be reporting mis-performance. Wrong-doers will be obliged to return part or all of the subsidy amount if they do not obey to the provisions in the signed contracts.

At least 100 households will be included in the programme and receive financial assistance/subsidy to decrease their emissions.

This sub-component will achieve the envisioned output of demonstrating the concept of **low emission district** by closely monitoring indoor and outdoor pollution levels, as well as energy savings over a 2-year period after investment. This will be achieved by providing households with monitoring tools and by encouraging them in recording and updating the platform (developed in activity 1.1.2) with their indoor pollution level and with data about heating practices and expenses. In coordination with the other monitor activities planned for Sub-component 1.1, outdoor pollution in the selected areas will be closely monitored to demonstrate emission reductions achieved.

Activity 3.1.5 Redesigning the model of municipal subsidies for the most the vulnerable categories of population

This sub-component aims to redesign the current "first come, first served" subsidy models that the Ministry of Economy and municipalities use by tailor-making subsidies for a) better off families to replace their inefficient stoves, and b) families with low income.

The current subsidy system has several shortcomings: 1) it is mostly used by households that have higher income and can bear the upfront payment; 2) the subsidies are not targeted to one specific group or area thus it is impossible to measure the impact; 3) it might be easily corrupted.

Within this activity, series of co-design workshops will be conducted with the key counterparts and users, such as: the Ministry of Economy, the Ministry of Labour and Social Policy, Ministry of Environment and Physical Planning, City of Skopje and all interested municipalities, interested citizens. CSOs, retailers and other private companies.

At the end of the project, the models designed and tested will be presented to the Intersectoral Group on Air Pollution and the Municipal Councils of the Skopje agglomeration.

The key expected results of Component 3 are:

1. Functional model for building of low-emission districts and later low-carbon cities.
2. Subsidy model for low-income families to replace the current first-come first-served models of the City and municipalities which does not give desired results and cannot be used by low-income families
3. Model for provision of subsidies for better off families to replace their inefficient stoves with solar systems or other less-polluting technologies, through empowering retailers to design and offer affordable and attractive packages.

Component 4: Build public awareness

Previously conducted studies highlighted the low level of awareness among the public about sources and effects of air pollutions, as well insufficient knowledge of the effective means to tackle the problem. The later also applies to number of CSOs which prevents them to participate in a meaningful dialogue with the Government, and to constructively contribute to identification of the solution and implementation of most appropriate actions and measures. Media also has a role to play by correct presentation of facts, sharing good but also bad practices, raising awareness and contributing to increase of knowledge. This component addresses the need to inform and engage citizens of all ages in 1) the pilot area and 2) in whole Skopje and other cities with high pollution, about the actions that need to be taken to shift behaviours that affect pollution levels.

Activity 4.1. Identification of target audiences, barriers and level of awareness, and desired attitude change

These short-term communications activities will be linked to Component 3 and focus on demonstrating the benefits in terms of quantified emissions reductions, savings and indirect benefits (e.g. health). First, this sub-component will identify target audiences, their barriers and level of awareness, and desired attitude change through a marketing survey. Moreover, capacity gaps of the NGOs in terms knowledge, skills, strengths, weaknesses, opportunities, threats, assets and other elements to contribute to the air pollution problems solving will be identified. Also, capacity for change of the communities in the pilot area shall be assessed in terms of: sense of community; communication; participation; leadership; resources, knowledge, and skills; and ongoing learning. Media/Journalists will be the third domain of interest for the identification of gaps to fulfil their mission in the context of air pollution.

Activity 4.2. Co-design and implementation of behavioural-changing activities with the interested stakeholders

After the identification of target audiences, a series of behavioural-changing activities co-designed with the interested stakeholders will be carried out. Particular attention will be given to apply behavioural science to address polluting and environmental-degrading practices such as garbage burning and illegal wood supply. Identified interventions will be then implemented into effective and communication activities (e.g. interactive maps) and community-scale awareness measures.
For example, drawing conclusions from the outcomes of component 3, targeted outreach and awareness activities will be designed with the purpose of addressing perceptions about individual contributions to the pollution.

Based on the findings of the capacity/gap assessments under the Activity 4.1, targeted capacity building programes for the NGOs, communities and media shall be designed and implemented. Among the other, these programmes shall capacitate them to actively contribute to decision-making process, engage in policy dialogue and monitor the implementation of the relevant polices/plans/programmes.

During the inception phase of the project all CSOs will be invited on a coordination meeting aimed at determining their interest to participate at the project, particularly in the implementation of this project component. This meeting shall be used to identify other ongoing and planned air pollution public awareness/advocacy activities carry out by CSOs with an aim to create synergies and to speak with one voice. Moreover, at this meeting CSOs will nominate their representative at the Project Board.

Activity 4.3 Transfer of Swedish outstanding residential low-emissions and sustainable living practices in the country.

The project will seek to identify and bring the best models to raise awareness and increase the knowledge among the key stakeholders. This will be focused but not limited to educational programmes, voluntary schemes for energy auditors and advisers, CSOs, private sector.

It is crucial that interventions under this Component are closely monitored and evaluated in order to understand which the most effective channels are, to stimulate behavioural changes, and which target communities can act as championing examples of low-emission districts.

Partnerships

A systematic approach to communication and awareness-raising will be applied to mobilize additional stakeholders and resources, and to extend partnerships for the implementation of the key project activities.

As part of its attempt to broaden environmental awareness beyond policy circles, UNDP will also deepen its ties with civil society organizations and research and academic institutions. Moreover, UNDP will ensure close collaboration and input from other relevant UN agencies that are part of the Working Group on Air Pollution which is established as part of the UN Country Team. In this context, important will be collaboration with the WHO due to the important aspects of the health issues related to air pollution, and UNWOMAN for the gender aspects.

Close coordination will be established with other international organisations in the country that are funding projects of similar scope and nature. Moreover, synergies will be sought with other ongoing UNDP projects where the City of Skopje is a partner and or a beneficiary.

Other projects which have been identified as having the greatest synergetic potential are:

- Cooperation project between the **Swedish Environmental Protection Agency** and the Ministry of Environment and Physical Planning (MoEPP) with the overall objective to support the preparation for EU accession in the environmental sector with particular focus on air pollution. This project supports the following activities: Upgrade of the currently established AIRVIRO software for air quality data management including air quality dispersion module; Upgrade the established e-reporting system for air quality, according to the EU Decision 2011/850/EU with new processes for air quality assessment, zones and agglomeration, plans, measures and air quality improvement scenarios; Replacement of outdated air quality measurement instruments included in the State air quality monitoring network; Training on using the air quality dispersion modelling tool in conducting air quality assessment studies; Establishment of National Emission Data Collection System (NEDCS) including establishment of a comprehensive National Cadastre of air emission polluters and strengthening emission measurement capacities.
- **EBRD** is currently active in the country with three related initiatives - Regional Energy Efficiency Programme Phase II and Green City Action Plan development (planned for Skopje). The EBRD has also launched a Green Economy Financing Facility (GEFF) that will provide loans to households for energy efficiency improvements in their homes. Ohridska Banka Société Générale and Sparkasse Bank Makedonija have signed up as the first two local partner financial institutions, with credit lines of €2 million each for on-lending to homeowners. An additional €4 million in financing is about to be committed with other

local banks. These investments will help households to become more energy efficient, decrease their greenhouse gas emissions and improve their living standards. The GEFF program in the country will support all households with a grant component of up to 20% of the value of the loan that will invest in: thermal insulation of walls, floors or roofs; windows and doors; a boiler or a pellet stove; heat pumps; solar collectors for hot water; photovoltaic panels for the production of electricity; lighting; ventilation and storage boilers for hot water.

- The Green Economy Financing Facility (GEFF) in the country is implemented under the **Regional Energy Efficiency Programme for the Western Balkans (REEP Plus),** funded by the **EU** and implemented in partnership with the Energy Community Secretariat. The programme also supports the development of the country's Energy Efficiency Law in compliance with the European Directives on energy efficiency and performance in buildings. In total, the REEP Plus programme will benefit from €135 million for investments in the residential sector, supported by €30 million in EU funding and €1.8 million from the European Western Balkans Joint Fund, as well as €3.3 million provided by the Austrian Federal Ministry of Finance.
- **USAID** has been involved in supporting energy efficiency in multi-apartment buildings especially focused on issues of fuel poverty via Habitat for Humanity and is supporting the development of a new EE law.
- **UNDP** is implementing a project Resilient Skopje: Scaling-up for Sustainability, Innovation and Climate Change. The long-term objective of the project is to assist the City of Skopje to become more resilient to climate change and other environmental challenges, and to build a public administration that will design and deliver innovative and more efficient services for its citizens.
- **UNDP**'s project "If we were counting" funded through the City Experiment Fund. "If We Were Counting" is a real-time governance test case that explores what could happen if we use algorithms to automatically regulate and fine heavy polluters and give that money back to residents to decide how to use it toward improving livelihoods in Skopje. This will be done through three experiments which jointly address the multiple dimensions of the challenge, and it aims to incentivize government and industry to work on new solutions to decrease air pollution, bring transparency to corrupt processes, decrease apathy, and increase trust in government.

Risks and Assumptions

The key assumptions that will underpin the project's success include:

- The Government puts the air pollution issues higher on the development agenda of the country and allocates enough financial resources for implementation of relevant strategies and action plans (eg. Clean Air Plan).
- The country stays committed to the European Integration path, and continues to approximate its legislation with relevant EU Directives;
- The Ministry of Environment and Physical Panning ensures regular monitoring of air quality, makes data and information publicly available, and commits to the expansion of the air monitoring network;
- Public awareness and understanding of sustainable development are increased and NGOs and citizens are more interested to actively contribute by decreasing their ecological footprint.
- City of Skopje and municipalities of the Skopje agglomeration demonstrate a strong interest and commitment to take over more responsibilities in the field of environment protection and allocate adequate resources (financial and human).
- CSOs demonstrate a strong interest and commitment to take on more responsibilities in the field of environment protection in close collaboration and partnership with central and local governments.

- National institutions at central and local level are committed to considering and planning development interventions in the areas of environment protection to meet the gender different needs and priorities.
- National and local governments are opened to new ideas, initiatives and partnerships, and to changing the way of planning and distributing subsidies.
- Private sector is interested to create partnerships with central and local government and with the citizens.

The implementation of this project depends on the appropriate identification and mitigation of potential risks which undermine a successful achievement of the project results. There are no major social and environmental risks that can threaten the achievement of the project results. The main external risk for the project delivery relates to the political situation in the country and extraordinary and/or regular parliamentarian elections, which are expected to happen during the lifetime of the project implementation. In order to mitigate this risk, the project team will maintain a non-partisan stance, and its focus on the mission of bringing tangible project results while standing ready to respond to possible shifts in national priorities.

The table below provides a summary of the key risks for this project, as well as the mitigation activities to avoid and manage them.

Risk	**Level of criticality**	**Likelihood of materializing**	**Explanation and mitigation activity**
Political risk – stalled support for clean air / EU agenda	Medium	Medium	The country is firmly on the path towards Europe and approximation of the laws with the relevant EU Directives. However, there is a risk that the adoption of the regulatory framework to encourage efficient appliances / DH connection could be delayed if there is no sense of urgency. To mitigate against this risk, the project will engage closely with the Government and municipality - explaining the benefits of the project and developing a regulatory impact assessment to examine the costs and benefits. Additionally, public awareness raising programmes and close coordination with other donors that support the EU agenda, particularly EU shall contribute assist in mitigation of this risk.
Parliamentarian elections announced for early spring might slow down the implementation of the project in the first half of 2020	Medium	High	The work plan for 2020 shall be drafted having in mind potential impact of the elections, and the most critical activities that require involvement of the central and/or local governments shall be planned by the end of March and after May 2020.

Insufficient national capacities to plan, design and implement an integrated approach for talking the air pollution that are result- and impact-oriented and will ensure sustainability.	High	Medium	UNDP in coordination with other interested donors and partners will make efforts to keep the air pollution high on the agenda both at central and local level by advocating linking those issues directly to the country's sustainable development. In addition, UNDP will help government to unlock the internal capacities in the government, in the private sector and in civil society, through multi-agency processes and facilitated partnerships with various stakeholders.
Public perception, insufficient interest, or push-back for the new concepts instigated by the project	Medium	Medium	The public perception of the project and its undertakings will be important in that there could be a lack of interest in the financing mechanisms, or perhaps even push-back against new concepts. To mitigate against this risk, it will be critical to conduct public awareness activities as described in Component 4 as well as ensure that proper support mechanisms are put into place to ensure that poorer households are not detrimentally impacted by the project's activities.
Corruption and political manipulation of subsidies	High	Medium	The subsidies are always a concern and subject to different political manipulations and assumptions, particularly reinforced during the pre-election period. To mitigate against this risk, all criteria and the methodology for the subsidies will be co-design with the end-users and transparently announced, while the selection of the neighbourhood will be done by UNDP in consultation with the key project partners.

In order to meet the below risks as well as to handle other developments, the management of the project will be flexible. The project management will be prepared throughout the project implementation to adjust the planning of activities and to reallocate resources. If deemed necessary, the scope reduced.

Stakeholder Engagement

Key stakeholders and roles

	Stakeholder	Role
1.	Ministry of Environment and Physical Planning	• air monitoring network maintenance and reporting; • protection of air, waters, soil, flora, fauna, and ozone; • restoration of polluted areas of environment; • proposing measures for solid waste management;

		• physical planning; • physical informative system; • supervision within its competencies;
2.	Ministry of Economy	• securing energy stability and safety; • development of energy policies, including renewables and energy efficiency strategies and plans, and development of respective laws; • providing subsidies for energy efficiency measures and installation of solar panels; • improvement of conditions for development of the industry; • clustering and the involvement of Macedonian companies in the network of global supply chains; • development of entrepreneurial sector, creation and integration of a single market in the EU,
3.	Ministry of Labour and Social Policy	• creation of the social policy of the country, with focus on the most vulnerable groups; • provision of financial assistance to the temporary unemployed people; • provision of different types of financial assistance for most vulnerable categories of the population, such are: families with no or very low income, care for persons with disabilities, elderly with illness, etc.; • creation and implementation of policies on mainstreaming gender equality; • social protection of children, youth, women and the people with disabilities;
4	Ministry of Finance	• creating stable public financing and stable macroeconomic framework; • providing an enabling environment for continuation of the process of economic reforms and acceleration of the economic growth; • contributing to better welfare and improvement of the living standard of the Macedonian citizens.
5.	City of Skopje/Municipalities of Skopje agglomeration	• protection of the environment, nature and space regulation: measures for protection and prevention of water, atmosphere and land pollution, protection of nature, protection against noise and ionizing radiation; • urban and rural planning: urban planning and issuing of technical documentation for construction and issuing construction permits; regulation and maintenance of construction land; • local economic development: local economic development planning; determining development and structural priorities; running the local economic policy; support for the development of small and medium-sized enterprises and entrepreneurship at local level and in that context participation in the establishment and development of a local network of institutions and agencies; promotion of partnership; • provision of subsidies for replacement of inefficient stoves; • potential direct and/or parallel funding; • complementary projects that will create synergies;
6	Intergovernmental Group on Air Pollution	• the overarching political platform, providing a high-level support for the development and realization of the activities related to mitigation of the air pollution • ensuring input and feedback from the respective ministries, private sector and CSOs that are part of the group

7	CSOs	• awareness raising, communication and education activities; • knowledgeable about environment protection issues; • partner in co-design of innovative solutions that will increase resilience and improve environment protection. • participating in experimenting and testing prototype services and solutions; • support the scaling the successful prototype solutions into larger projects
8	University "Goce Delcev", Stip	• responsible party for the monitoring component of the project • knowledge sharing
9	Academia	• knowledge sharing
10	Private sector	• potential partner for implementation of measures
11	Energy Auditors/ Association of Energy Auditors	• knowledge and experience on energy audits
12	Donors	• potential partners for expansion of the programme • technical knowledge and experience

To achieve greater impact in reducing local emissions, the activities will be directed towards the target group of households using coal and firewood in the municipalities whose share in firewood consumption is the largest, and also, in parts of municipalities which is not expected to be included in the expansion of the district heating network. Moreover, the main target group will be households with low monthly income which need to be assisted to change the heating technology or/or improve the house insulation.

The household heating survey identified potential target groups in each of the municipalities that are part of Skopje agglomeration, considering the following factors that are intertwined or complement each other, which in economic theory are considered to be crucial regarding the choice of way of heating:

1. Economic conditions – households' economic status as a determining factor that directly affects the choice of energy source.
2. Demographic and cultural features - household size, age structure, education level.
3. Insulation factors of residential buildings - ownership status, the building's year of construction and size.
4. Motivational factors - lifestyle habits of family members, their level of comfort, use of new techniques and technologies in the home, awareness and commitment to environmental protection and family members' health.

Having into consideration the availability of funding, it is proposed to focus the activities in the Municipality of Aerodrom i.e. the settlement Lisice. Due to the specific location of this settlement in the Skopje valley, the measurements of the air quality during the winter period show very high values for prolonged period. Individual houses are predominant in Lisice, and 65% of the households using wood that were surveyed do not have proper insulation. More that 80% of them have monthly family income below 30,000 MKD (less than 500 Euros), and 76% are families with children below 18 years.

The project team will create mechanisms for effective stakeholder engagement and will ensure the meaningful participation of targeted groups/geographic areas. Particular efforts will be made to take into consideration the needs of excluded and marginalized groups.

Various tools will be used for stakeholder engagement, including design thinking, behavioural science, and foresight. The project will create a collaborative space where citizens and local officials will join forces to design, prototype and test concrete actions that will improve the air quality on the City of Skopje thus improving the wellbeing of its citizens.

Compliance and response mechanisms

UNDP shall also ensure that potentially affected people have access to and are aware of mechanisms to submit concerns about the social and environmental impacts of a project. The key instruments which will be used are UNDP's Social and Environmental Compliance Review and Stakeholder Response Mechanism

(http://www.undp.org/content/undp/en/home/operations/accountability/secu-srm.html).

UNDP's Social and Environmental Standards (SES) underpin its commitment to mainstream social and environmental sustainability in its Programmes and Projects to support sustainable development. The objectives of the Social and Environmental Standards Procedure are to: (a) integrate the SES Overarching Principles (human rights, gender equality and environmental sustainability); (b) identify potential social and environmental risks and their significance; (c) determine the project's risk category (Low, Moderate, High); and (d) determine the level of social and environmental assessment and management required to address potential risks and impacts. The Social and Environmental Compliance Review is mandatory for all UNDP projects worth more than USD 500,000, and therefore the project must undergo this process.

The Stakeholder Response Mechanism (SRM), on the other hand, provides a supplemental, formal avenue for stakeholders to engage with UNDP. The SRM will be available to project-affected stakeholders, government agencies and other partners to jointly resolve concerns and disputes when they believe that the project may have adverse social or environmental impacts; they have raised their concerns with UNDP through standard channels for stakeholder consultation and engagement; and they have not been satisfied with the response. This mechanism can help the concerned parties to start or restart dialogue, facilitate discussions, mediate disputes, enhance understanding of the facts, and undertake other activities that might help resolve concerns and disputes.

Knowledge

The project will produce knowledge products for government, donors, private stakeholders and for the wider public.

Under Component 1, a better understanding of the causes of air pollution will be achieved through the upgrade of the current monitoring system with IoT and machine learning-based IT platform, through the citizen-led monitoring system, through the repetition of the heating study and through the source apportionment study. These activities will produce knowledge (in the forms of reports, data visualisation etc.) that will not only support investment and regulatory decision-makers, but it will also contribute to the awareness and communication activities, ultimately helping society to understand how their behaviours affect air quality.

Results from the monitoring and modelling activities will be communicated through the interactive platform where citizens will be able to understand data presented in a compelling and visually appealing way.

The activities envisioned under Component 4 will produce visibility and communication products that inform about the benefits achieved under Component 3, such as quantified emissions reductions, savings and indirect benefits (e.g. health). In addition, the interventions co-designed with stakeholders to address polluting and environmental-degrading practices will be implemented

into effective communication activities, such as interactive maps, videos and other community-scale awareness measures.

The educational and awareness raising programme will produce a number of modules which can be included into schools' curricula and can inspire actions for shifting towards a low-emission economy, as well as practical activities to encourage people of all ages to take action in monitoring air pollution.

In summary, the activities planned for Component 4 have the specific goal of producing insight-led campaigns and interventions that inform citizens and support them in taking the necessary steps to shift towards sustainable behaviours.

Furthermore, the project's monitoring and evaluation report will produce lessons learned so that other stakeholders can benefit.

Sustainability and Scaling Up

Sustainability is a core principle of the project strategy. The project design is based on a thorough analysis of the root causes of the air pollution in the City of Skopje, focusing on one of the main contributors – household heating. By addressing these issues, the project will contribute directly to overcoming some of the main constraints, thus improving overall sustainability prospects.

The project will work with a targeted group of beneficiaries and establish models for low emission districts. This area of the city shall become innovation model that is expected to scale up and expand the innovation spirit across the other municipalities that are part of Skopje agglomeration.

The high-impact demonstration activities envisioned for Component 3 will be designed with the purpose of validating proof of concept and they will seek active participation of communities and civil society organisations. This co-design approach ensures ownership of solutions in the long term, as well improves the capacity of actors involved to scale up initiatives.

Furthermore, the project is also designed to fill in the monitoring data and information gap necessary to make informed decision about possible investment in air pollution measures, as well as to test models that for talking the air pollution through investment in energy efficiency and subsidy models and as such have a great scale-up potential.

All activities that will be implemented are part of the Clean Air Plan which further ensures the ownership of the City of Skopje and the Government.

The project is also designed to enable significant stakeholder participation and promote broad awareness about the air pollution in the medium and long term. This is considered crucial to the social sustainability of all results.

A coordination platform led by UNDP will support the creation of strategic partnerships and the dissemination of lessons learned among stakeholders. This platform will build the basis for future partnerships and it will ensure that all relevant stakeholders are aligned in tackling the air pollutions problem in a comprehensive way.

IV. PROJECT MANAGEMENT

Project Office

Day-to-day implementation will be carried out by a project team composed of a Project Manager, a Project Assistant and a Communication Officer that will be shared with other projects in the environmental portfolio.

The Project team will be based in UNDP premises. The project office costs will include expenses for rent, electricity, heating, water, utilities, internet, security, cleaning and maintenance, telecommunication services, based on UNDP monthly average expenses for such services. The costs will be pro-rated. The project office costs shall also include: IT Equipment for the Project Manager and the Project Assistant (2 work stations, including laptops, docking station, monitors, printer/copier/scanner, based on UNDP estimations for similar assets); costs for office supplies for

18 months for the needs of the project office (printing/copying paper, office stationary); communication costs (mobile telephones, telephone services and e-mail subscription services for the project staff);

The Project Manager's prime responsibility is to ensure that the project produces the results (outputs) specified in the project document, to the required standard of quality and within the specified constraints of time and cost.

The Project Assistant will be responsible for the project administrative support including finance, procurement and general administration. S/he will implement assigned project components, develop project resource profiles and timelines, assess delivery issues and work with national counterparts to build common understanding and coordination on project activities.

The Communication Officer will support the implementation of the public awareness and communication activities and will ensure the proper visibility of the project results. The Communication Officer will also cultivate good working relationships with participating organizations and entities and ensure a regular exchange of information among all project partners.

A Chief Technical Advisor will be engaged on Individual Contract (consultancy) to provide technical support to the project team and partners as needed. S/he will communicate regularly with the relevant stakeholders and beneficiaries, provide advice as needed, and facilitate their involvement in the relevant activities. S/he will be expected to support the reporting and evaluation of implementation of the project activities.

UNDP DIRECT PROJECT COSTED STAFF AND ROLES

In addition to the project staff, depending on the nature of the work and complexity a number of technical and administrative roles and services are covered by the UNDP country office and are cost-shared within the office.

The Programme Officer in charge of the Energy, Environment and Disaster Risk Reduction Portfolio will provide strategic guidance, policy advice and technical input essential to deliver development results. She will also create synergies with other complementary interventions which contribute to the achievement of the overall project goal. She will oversee the project implementation, monitor and report to the donor, review and approve the TOR (programmatic aspects), serves as a Chair of the Evaluation Committee, and will facilitate decision making to ensure project implementation proceeds in a flexible but efficient manner. She will be the key focal point for coordination between the Project, UNDP, SIDA and the national partner authorities and other key Project stakeholders.

Programme and Monitoring & Evaluation Associate will ensure effective implementation of UNDP monitoring and evaluation policies; the alignment of the project with the UNDP monitoring framework; the introduction of monitoring and evaluation tools to enable data collection and analysis and will support the regular project reporting.

The UNDP Operations team will provide administrative support in terms of procurement, operations management, human resources, financial management, and other required administrative support.

The Operations Manager will be directly involved in procurement and HR processes related to project implementation in line with the SOP including but not limited to: providing inputs to TOR, specifications, endorsing procurement processes, recruitments and HR management for project needs, disbursement officer for payments. In addition, he will provide, quality assurance, advises on procurement and HR processes for the need of the project. He will manage external relations related to all operational aspects of the project.

The Procurement Associate will assist project implementation through facilitating quality, transparent, effective and fast procurement processes; reviewing and announcing procurement

processes; provide direct advisory support in procurement/tender evaluation processes; support in negotiations with potential contractors (as needed); assistance in the process of contracting, monitoring of contracts.

The Programme Finance Associate will provide support in preparation of the budget, budget revisions, support to overall financial monitoring and reporting for the overall action. He will assist the project team in preparation of financial transactions and appropriate financial reports.

UNDP will make sure that the project staff possess the necessary combination of skills and interdisciplinary expertise in response to the complexity of the project actions. The key staff involved in the project implementation will be compensated for their services in line with UNDP's rules and procedures and their contribution to the implementation of project activities.

Remuneration / indirect costs corresponding to 8% GMS which UNDP is mandated to recover in line with UNDP's Executive Board's and relate to the corporate level costs and embrace e.g. corporate executive management, corporate legal support, corporate legal management, policy guidelines on procurement and logistic support etc.

All other purchased equipment, and major supplies by necessary for the realization of the projects will be transferred to the respective beneficiaries. Matters relating to the transfer of ownership by UNDP to the national partners will be processed in accordance with the relevant policies and procedures of UNDP.

The project will be implemented in the period of 25 months which is considered as optimal for completion of all project activities.

V. RESULTS FRAMEWORK

Intended Outcome as stated in the UNDAF/Country Programme Results and Resource Framework:

4. By 2020, individuals, the private sector and state institutions base their actions on the principles of sustainable development, and communities are more resilient to disasters and environmental risks.

Outcome indicators as stated in the Country Programme Results and Resources Framework, including baseline and targets:

Applicable Output(s) from the UNDP Strategic Plan:

1.5.1: Solutions adopted to achieve universal access to clean, affordable and sustainable energy

2.5.1: Solutions developed, financed and applied at scale for energy efficiency and transformation to clean energy and zero-carbon development, for poverty eradication and structural transformation

Project title and Atlas Project Number: 00109164 Tackling Air Pollution in the City of Skopje

Contribution to SDGs targets achievements:

3.9. By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.
7.1. By 2030, ensure universal access to affordable, reliable and modern energy services
7.3. By 2030, double the global rate of improvement in energy efficiency
11.6 By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management
11.A Support positive economic, social and environmental links between urban, peri-urban and rural areas by strengthening national and regional development planning
13.3 Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning

EXPECTED OUTPUTS	OUTPUT INDICATORS	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)			DATA COLLECTION METHODS & RISKS
			Value	Year	Year 2020	Year 2021	FINAL	
Contribution to Agenda 2030 SDGs 1.6.2. Indicator	Annual mean levels of fine particulate matter (e.g. PM2.5 and PM10) in cities (population weighted)	*Air Quality Monitoring System of the Ministry of Environment and Physical Planning*	To be determined for Lisice	2019	5% decrease	10% decrease	10% decrease (cumulative)	*Official air quality monitoring system of MoEPP* *WHO to be consulted*
Contribution to UNDP Strategic Results Framework Solutions adopted to achieve universal access to clean, affordable and sustainable energy	*Share of households benefitting from clean, affordable and sustainable energy access in Skopje:* *a) Women-headed* *b) Skopje* *c) Aerodrom*	*Database on households purchasing efficient / clean stoves - disaggregated by type of household and household heating practices survey 2020*	TBD	2019	TBD	TBD	TBD	*Perception survey, 2020* *No risks are associated with measuring of this indicator*
Contribution to UNDP Strategic Results Framework	*Volume of investment leveraged from public and private sources through UNDP support for national*	*Ratio of Additional investment leveraged from companies, private households for*	0	2019	/	1:1	1:1	*Reporting from grant disbursements and from partner*

Solutions developed, financed and applied at scale for energy efficiency and transformation to clean energy and zero-carbon development, for poverty eradication and structural transformation	*programmes/initiatives*	*stoves, private households for building efficiency, and from Government / municipal sources*						*organisations* *Risks: Difficulties to secure the funding from the partnering organizations, possibility to disburse their funds to different locations*
	Solutions applied at scale to accelerate transition to improved energy efficiency and clean energy	*Number of additional people impacted by energy efficiency / clean energy investments (disaggregated by gender)*	0	2019	/	3	3	*Tracking forms for grant disbursements and reporting from DH companies*
	Decrease of air pollution levels in the selected area (in percentage)	*Air quality monitoring stations installed around the selected area*	Tbc (the baseline level of pollution will be determined between December 2019 – April 2020, before start of measures implementation)	2019/2020	/	30% decrease	30% decrease	*Monitoring equipment, laboratory analysis*
	Decrease of polluting emissions from the selected area (in percentage)	*Calculation of emissions according to international standards*	Tbc (the baseline will be calculated with the energy audits)	2019	/	70% decrease	70% decrease	*Formulas, mathematical calculations per internationally accepted standards*
Output 1.1 Comprehensive monitoring system for the pilot area developed Gender marker: Disaggregated information on gender impacts of air pollution	*Air pollution monitoring systems in place, Monitoring and online platform developed, Study developed with gender disaggregated data on exposure*	*Existence of outputs*	0	2019	Air pollution monitoring systems in place	Monitoring and online platform developed	Study carried out and all other outputs achieved	*Terminal evaluation assessing the outputs, periodic project reports*

Output 1.2 Modelling of air pollution sources completed	*Source apportionment model developed*	*Existence of model and report describing outputs*	0	2019	N/A	Model developed and report completed	Model developed and report completed	*Terminal evaluation assessing the outputs, periodic project reports*
Output 1.3 Coordination platform implemented	*Number of stakeholder meetings convened*	*Meeting notes and materials*	0	2019	3	6	8	*Terminal evaluation assessing the outputs, periodic project reports*
Output 2.1: Legislation changes supported Gender marker: Analysis of impacts on gender	*Number of regulatory documents drafted and submitted to the appropriate authorities*	*Draft documents*	0	2019	2	3	5	*Terminal evaluation assessing the outputs, periodic project reports*
Output 3.2 Low emission district concept demonstrated Gender marker: Disaggregated indicator on individuals benefitting	*Number of households implementing changes (EE in house, heating technologies, waste management, greenery,*	*Contracts signed*	0	2019	-	100	100	*Reporting on grant disbursement.*
Output 4.1 Communication and visibility activities designed and implemented	*Marketing survey and co-design of awareness raising activities*	*Results of market survey and awareness raising activities planned / carried out*	0	2019	N/A	Market survey carried out and co-design of awareness raising activities	Market survey carried out and co-design of awareness raising activities	*Reports from implementing parties*
	Capacity/gap assessments and capacity building programmes	*Assessments and CB programmes*	0	2019	3 Assessments and 3 Programmes	-	3 Assessments and 3 Programmes	*Terminal evaluation assessing the outputs, periodic project reports*

VI. Monitoring And Evaluation

In accordance with UNDP's programming policies and procedures, the project will be monitored through the following monitoring and evaluation plans:

Monitoring Plan

Monitoring Activity	Purpose	Frequency	Expected Action	Partners (if joint)	Cost (if any)
Track results progress	Progress data against the results indicators in the RRF will be collected and analysed to assess the progress of the project in achieving the agreed outputs.	Quarterly, or in the frequency required for each indicator.	Slower than expected progress will be addressed by project management.		
Monitor and Manage Risk	Identify specific risks that may threaten achievement of intended results. Identify and monitor risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per UNDP's Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage financial risk.	At least annually	Risks are identified by project management and actions are taken to manage risk. The risk log is actively maintained to keep track of identified risks and actions taken.		
Learn	Knowledge, good practices and lessons will be captured regularly, as well as actively sourced from other projects and partners and integrated back into the project.	At least annually	Relevant lessons are captured by the project team and used to inform management decisions.		
Annual Project Quality Assurance	The quality of the project will be assessed against UNDP's quality standards to identify project strengths and weaknesses and to inform management decision making to improve the project.	Annually	Areas of strength and weakness will be reviewed by project management and used to inform decisions to improve project performance.		
Review and Make Course Corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	At least annually	Performance data, risks, lessons and quality will be discussed by the project board and used to make course corrections.		
Project Report	A progress report will be presented to the	Annually, and at			

	Project Board and key stakeholders, consisting of progress data showing the results achieved against pre-defined annual targets at the output level, the annual project quality rating summary, an updated risk long with mitigation measures, and any evaluation or review reports prepared over the period.	the end of the project (final report)			
Project Review (Project Board)	The Project Board will hold regular project reviews to assess the performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the project's final year, the Project Board shall hold an end-of project review to capture lessons learned and discuss opportunities for scaling up and to socialize project results and lessons learned with relevant audiences.	At least once a year	Any quality concerns or slower than expected progress should be discussed by the project board and management actions agreed to address the issues identified.		

VII. Multi-Year Work Plan

Outputs	Activities	Planned Budget by Year			RESPONSIBLE PARTY	PLANNED BUDGET		
		2019	2020	2021		Funding Source	Budget Description	Amount in USD
Component 1: Develop a comprehensive monitoring system for the pilot area and a coordination platform to tackle air pollution								
Sub-component 1.1: Development of a comprehensive monitoring system for the pilot area	1.1.1 Improvements to the existing monitoring system to measure air pollution by extending the network of monitoring stations and setting up an IoT and machine learning-based IT platform		$ 80,000		Ministry of Environment and Physical Planning	SIDA	71300 Local Consultants 72100 Contractual services - Companies 72300 Materials and Goods 72800 Information Technology equipment 74200 Audio Visual & Print Prod Costs 75700 Training, workshop, conference	$ 80,000
	1.1.2 Develop a monitoring system and an online platform for collecting and visualizing data		$ 15,000	$ 15,000	Ministry of Environment and Physical Planning			$ 30,000
	1.1.3 Continued monitoring of the contributions to the air pollution by repeating heating study at the end of the project to include both		$ 20,000		UNDP			$ 20,000

	small companies and households							
Sub-component 1.2: Modelling of air pollution sources	1.2.1 Source apportionment modelling - Identify specific geographical areas in which investing in mitigation measures will have the greatest impact		$ 80,000	$ 100,000	Stip University			$ 180,000
Sub-component 1.3: Implementation of a coordination platform	1.3.1 Learning exchange and linkages with other initiatives to inform a long-term plan of action - Convening stakeholders, donors and relevant actors to implement necessary changes identified		$ 17,000	$ 3,000	UNDP			$ 20,000
Sub-Total for Component 1			**$ 212,000**	**$ 118,000**				**$ 330,000**
Component 2: Implement regulatory changes necessary to transitions towards a lower emission household energy system								
Output 2.1: Legislation changes supported	2.1.1 Assistance to support legislation changes - eco-design adoption and		$ 10,000	$ 5,000	UNDP	SIDA	71300 Local Consultants 72100 Contractual services - Companies	$ 15,000

	implementation to eliminate poorly performing wood stoves / solid fuel heaters							
	2.1.2. Assistance to support legislation changes - waste management (promote recycling of plastic)		$ 5,000					$ 5,000
	2.1.3. Regulatory impact assessment - cost-benefit, social, environmental, and gender impact for regulatory changes			$ 25,000				$ 25,000
Sub-Total for Component 2			**$ 15,000**	**$ 30,000**				**$ 45,000**
Component 3: Demonstration of measures that address the causes of pollution for household heating to show proof of concept								
Output 3.1 Subsidy programme for low-emission energy efficient districts	3.1.1. Establishment of partnerships with all relevant stakeholders from private and civic sectors and formalization of the partnership with the Municipality of Aerodrom and		$ 5,000		UNDP	SIDA	71200 International Consultants 71300 Local Consultants 72100 Contractual services - Companies 75700 Training, workshop, conference	$ 5,000

	Gjorce Petrov							
	3.1.2 Selection of intervention neighbourhoods		$ 5,000	$ 10,000	UNDP			$ 15,000
	3.1.3. Energy audits and co-design of energy-efficient measures, less-polluting heating systems and yard/area improvements with selected stakeholders		$ 30,000	$ 10,000	UNDP			$ 40,000
	3.1.4 Financing of interventions: subsidy for thermal rehabilitation for individual houses combined with redesign / installation of new heating elements, rubbish removal and yards arrangements where needed		$ 352,523.67	$ 650,000	UNDP			$ 1,002,523.67
	3.1.5 Redesigning the model of municipal subsidies for the most the		$ 10,000	$ 4,000				$ 14,000

	vulnerable categories of population							
Sub-Total for Component 3			**$ 402,523.67**	**$ 674,000**				**$ 1,076,523.67**
Component 4: Build public awareness								
4. Build public awareness	4.1. Identification of target audiences, barriers and level of awareness, and desired attitude change		$ 28,000		UNDP via a CSO	SIDA	71300 Local Consultants 71600 Travel 72100 Contractual services - Companies 74200 Audio Visual & Print Prod Costs 75700 Training, workshop, conference	$ 28,000
	4.2. Co-design and implementation of behavioral-changing activities with the interested stakeholders		$ 46,000	$ 66,000	UNDP via a CSO			$ 112,000
	4.3. Transfer of Swedish outstanding residential low-emissions and sustainable living practices in the country.		$ 30,000	$ 30,000				$ 60,000
Sub-Total for Output 4			**$ 104,000**	**$ 96,000**				**$ 200,000**
Evaluation								
Final Evaluation	EVALUATION	-	0	$ 20,000	UNDP (via external expert)	SIDA	74100 Professional Services	$ 20,000
Sub-Total for Evaluation		**-**	**-**	**$ 20,000**				**$ 20,000**
Project Management								
Project Management Unit		$ 5,177	$ 62,124	$ 62,124	UNDP	SIDA	71400 Service Contract	$ 129,425

		$ 2,634	$ 31,608	$ 31,608			64397 Direct project costs – Staff	$ 65,850
		$ 7,263	$ 15,176	$ 15,176			74596 Direct Project Costs – General Operating Expenses (GOE)	$ 37,615
Sub-Total for Project Management		**$ 15,074**	**$ 108,908**	**$ 108,908**				**$ 232,890**
General Management Support		$ 1,205.92	$ 67,394.53	$ 83,752.64				$ 152,353.09
Total project budget		**$ 16,279.92**	**$ 909,826.20**	**$ 1,130,660.64**				**$ 2,056,766.76**

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

The project will be executed by UNDP Country Office under the Direct Implementation Modality (DIM) in line with UNDP's Programme and Operations Policies and Procedures. UNDP will be responsible for overall management, ensuring project coherence, the preparation and implementation of work plans and annual audit plans; preparation and operation of budgets and budget revisions; disbursement and administration of funds; recruitment of national and international consultants and personnel; financial and progress reporting; and monitoring and evaluation.

For the delivery of specific activities related to the air quality monitoring component, the University "Goce Decev", Shtip will be engaged as a Responsible Party for the Project. The University has been previously engaged by the City of Skopje to carry out studies for air pollution, expanding the monitoring network of monitoring stations etc, and has collaborated with UNDP for the heating study for Skopje agglomeration. They are also part of the newly established working group for systemic usage of scientific data for evidence - based policy making on air pollution.

Than years ago, the University in Shtip established an AMBICON Lab as a part of the Faculty of Natural and Technical Science as an entity focused on development of methodological base for ambient air monitoring and analysis. In the current format, AMBICON unites three scientific groups: a) ambient control and sampling group; b) electrons scanning microscopy group, and c) elemental analysis group.

Since 2014, the Lab is accredited in accordance with the ISO 17025 for environment and samples from the environment testing areas, including but not limited to the following methods:

- standard gravimetric measurement method for the determination of the PM10/PM2,5 mass concentration of suspended particulate matter (ISO 12341:2014);
- determination of chemical composition of ambient particulate matter using Energy Dispersive X-Ray Fluorescence-EDXRF (EPA/625/R-96/010a, Method IO-3.3);
- direct on filter analyses of ambient air particulates mineralogy by X-ray diffractometer – XRD (MDHS101/2:2014);
- particulate geometry/morphology and composition using Electron Scanning Microscopy equipped energy dispersive spectrometer (EDS);
- optical determination of Black Carbon and Aromatic Organic Compounds in ambient particulates by dual wavelength transmissometer.

Recently, a new five-year extension of the accreditation was granted to AMBICON (until 2024) with additional analytic capabilities and improved precision for all methods within testing area. Focus on improvement and innovation provide strong methodology standpoint for indicating possible sources through aerosols characteristics determination, as much as a using data as input for source apportionment studies.

In addition, AMBICON operates an independent monitoring network with in-house developed ambient particulate monitors. The monitors are based on near equivalent light-scattering photometry with IR LED and equipped with sharp cut cyclone and sample heater, thus largely extending the Lab testing capacity. In this context, the Lab collaborates on non-commercial bases with the Ministry of Environment and Physical Planning and local communities with an aim to increase scientific database and support monitoring efforts in areas with no coverage or specific air quality problems. Data are in real time shared with the public using the Lab portal and popular air quality applications.

DIM implementation modality has been chosen due to the specific scope of the project activities and the present situation in the country. Namely, the project activities will require high level of coordination and collaboration between many relevant stakeholders on national and local level such as: the Ministry of Environment and Physical Planning, the Ministry of Economy, the Ministry of Labour and Social Policy, the Ministry of Finance, the City of Skopje and the municipalities of Skopje agglomeration, Academia and CSOs. UNDP as an independent and neutral entity is well positioned to bring together all partners and to build consensus on such important topic that requires active participations of the national and local governments, civil society organizations and academia.

Moreover, the country is expecting extraordinary elections in April 2020. The elections could cause postponement of the implementation of the activities and timely competition of the project if the responsibility lies with the Ministry of Environment and Physical Planning or the City of Skopje.

Nevertheless, while implementing the project, UNDP will make particular efforts to ensure future sustainability, and shall work out measures to build up the national capacities as part of the project implementation. Also, UNDP will establish channels for public participation and will involve representatives of the key project partner in the decision-making processes.

The Inter-sectorial Working Group established by the Government will be the overarching political platform, providing a high-level support for the development and realization of the activities related to mitigation of the air pollution. It comprises of the key stakeholders from national institutions, research and educational entities, private sector and civil society, and is charred by the Deputy Minister of Environment and Physical Planning. The Inter-sectorial Working Group will also provide policy guidance to the project strengthening the inter-institutional coordination on the state of ambient air, identification of appropriate measures and actions for improvement of the air quality, from all sectors, such are: household heating, public transport construction works, industrial pollution, better waste management and awareness raising.

A Project Board will be established as the main decision-making body for the project responsible for making by consensus management decisions for a project. The Project Board decisions shell be made in accordance to standards that shall ensure best value to money, fairness, integrity transparency and effective international competition.

The Project Board is responsible for the following: provides overall guidance and direction to the project, ensuring it remains within any specified constraints, provides guidance and agrees on possible countermeasures/management actions to address specific risks; appraises the Project Annual Review Report and make recommendations for the next AWP; reviews and approves end project report, make recommendations for follow-on actions; provide ad-hoc direction and advice for exception situations; assess and decide on project changes through revisions; assures that all Project deliverables have been produced satisfactorily; reviews and approves the Final Project Review Report, including Lessons-learned.

The Project Board will meet at least once a year and could have ad-hock meetings in the course of the project implementation, if deemed necessary.

The composition of the Project Board will be the following:

Executive: The Executive is an individual who represents ownership of the project who will chair the Project Board. This role will be held by a high representative from UNDP (Deputy Resident Representative).

The Executive is ultimately responsible for the project, supported by the Senior Beneficiary and Senior Supplier. The Executive's role is to ensure that the project is focused throughout its life cycle on achieving its objectives and delivering outputs that will contribute to higher-level outcomes. The executive should ensure that the project gives value for money, ensuring cost-conscious approach to the project, balancing the demands of beneficiary and supplier.

Specific Responsibilities: (as part of the above responsibilities for the Project Board)

- Ensure that there is a coherent project organisation structure and logical set of plans;
- Set tolerances in the AWP and other plans as required for the Project Manager;
- Monitor and control the progress of the project at a strategic level;
- Ensure that risks are being tracked and mitigated as effectively as possible;
- Brief relevant stakeholders about project progress;
- Organise and chair Project Board meetings.

Senior Supplier: The Senior Supplier is an individual or group representing the interests of the parties concerned which provide funding and/or technical expertise to the project (designing, developing, facilitating, procuring, implementing). The Senior Supplier's primary function within the Board is to provide guidance regarding the technical feasibility of the project. A representative of Swedish SIDA will have the role of a Senior Supplier as the main provider of the project funding.

Specific Responsibilities (as part of the above responsibilities for the Project Board)

- Make sure that progress towards the outputs remains consistent from the supplier perspective;
- Promote and maintain focus on the expected project output(s) from the point of view of supplier management;
- Ensure that the supplier resources required for the project are made available;
- Contribute supplier opinions on Project Board decisions on whether to implement recommendations on proposed changes;
- Arbitrate on, and ensure resolution of, any supplier priority or resource conflicts.

Senior Beneficiary: The Senior Beneficiary is an individual or group of individuals representing the interests of those who will ultimately benefit from the project. The Senior Beneficiary's primary function within the Board is to ensure the realization of project results from the perspective of project beneficiaries. The Senior Beneficiary role will be held by the senior management representative from the Ministry of Environment and Physical Planning (Minister, Deputy Minister or State Secretary), the Mayor of the City of Skopje, and representative of CSOs.

The representative of the CSOs will be selected in a consultative process with the CSOs that are actively involved on awareness raising on the air pollution issues in the City of Skopje. The CSO representative should particularly ensure that the voice of the main target groups is heard and that they influence the decision making for the project.

The Mayor of Aerodrom will be invited as an observer.

The Senior Beneficiary is responsible for validating the needs and for monitoring that the solution will meet those needs within the constraints of the project. The Senior Beneficiary role monitors progress against targets and quality criteria.

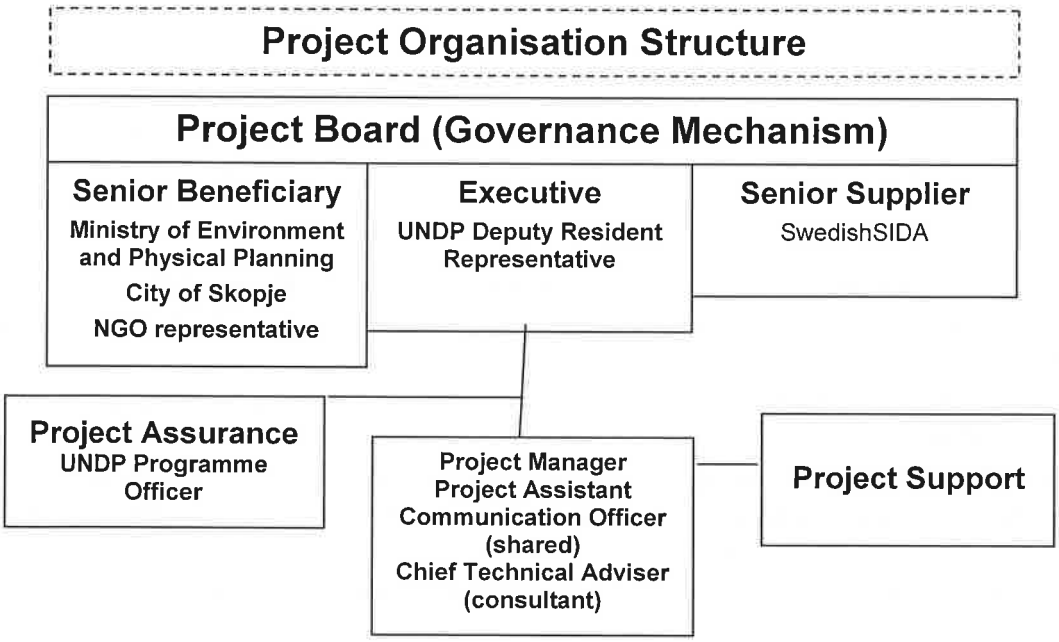
Specific Responsibilities (as part of the above responsibilities for the Project Board)

- Prioritize and contribute beneficiaries' opinions on Project Board decisions on whether to implement recommendations on proposed changes;
- Specification of the Beneficiary's needs is accurate, complete and unambiguous;
- Implementation of activities at all stages is monitored to ensure that they will meet the beneficiary's needs and are progressing towards that target;

- Impact of potential changes is evaluated from the beneficiary point of view;
- Risks to the beneficiaries are frequently monitored.

Project Assurance: UNDP provides a three – tier supervision, oversight and quality assurance role. The quality assurance role supports the Project Board and Project Management Unit by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. This project oversight and quality assurance role is covered by Programme Officer, Head of Energy & Environment Unit in UNDP Country Office.

The project organisation structure is as follows:



IX. LEGAL CONTEXT

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of fyr of Macedonia and UNDP, signed on 30 October 1995. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by UNDP ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures.

X. RISK MANAGEMENT

1. UNDP as the Implementing Partner will comply with the policies, procedures and practices of the United Nations Security Management System (UNSMS.)

2. UNDP as the Implementing Partner will undertake all reasonable efforts to ensure that none of the project funds are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

3. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

4. UNDP as the Implementing Partner will: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

5. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

6. UNDP as the Implementing Partner will ensure that the following obligations are binding on each responsible party, subcontractor and sub-recipient:

    a. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of each responsible party, subcontractor and sub-recipient and its personnel and property, and of UNDP's property in such responsible party's, subcontractor's and sub-recipient's custody, rests with such responsible party, subcontractor and sub-recipient. To this end, each responsible party, subcontractor and sub-recipient shall:
        i. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;

ii. assume all risks and liabilities related to such responsible party's, subcontractor's and sub-recipient's security, and the full implementation of the security plan.

b. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the responsible party's, subcontractor's and sub-recipient's obligations under this Project Document.

c. Each responsible party, subcontractor and sub-recipient will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, subcontractors and sub-recipients in implementing the project or programme or using the UNDP funds. It will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

d. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to each responsible party, subcontractor and sub-recipient: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. Each responsible party, subcontractor and sub-recipient agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

e. In the event that an investigation is required, UNDP will conduct investigations relating to any aspect of UNDP programmes and projects. Each responsible party, subcontractor and sub-recipient will provide its full cooperation, including making available personnel, relevant documentation, and granting access to its (and its consultants', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with it to find a solution.

f. Each responsible party, subcontractor and sub-recipient will promptly inform UNDP as the Implementing Partner in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

Where it becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, each responsible party, subcontractor and sub-recipient will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). It will provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

UNDP will be entitled to a refund from the responsible party, subcontractor or sub-recipient of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of this Project Document. Such amount may be deducted by UNDP from any payment due to the responsible party, subcontractor or sub-recipient under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail any responsible party's, subcontractor's or sub-recipient's obligations under this Project Document.

Where such funds have not been refunded to UNDP, the responsible party, subcontractor or sub-recipient agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to such responsible party, subcontractor or sub-recipient for the recovery of any funds determined by

UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

g. Each contract issued by the responsible party, subcontractor or sub-recipient in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from it shall cooperate with any and all investigations and post-payment audits.

h. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project or programme, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

i. Each responsible party, subcontractor and sub-recipient shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to its subcontractors and sub-recipients and that all the clauses under this section entitled "Risk Management Standard Clauses" are adequately reflected, *mutatis mutandis*, in all its sub-contracts or sub-agreements entered into further to this Project Document.